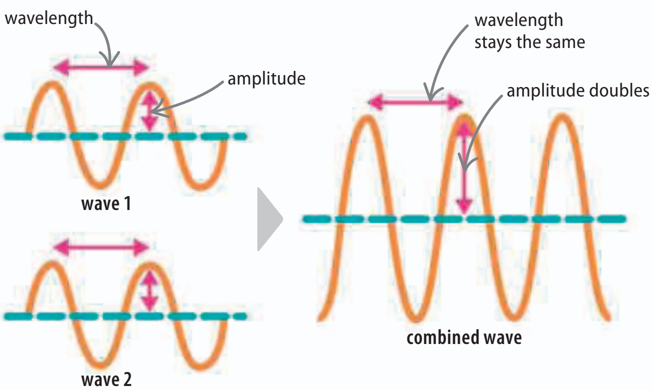
Interference

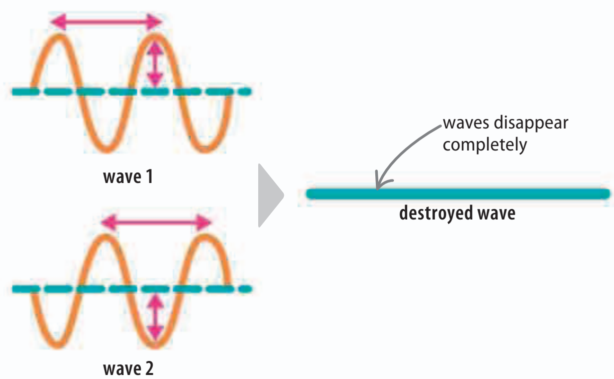
Where two rays of light meet, they affect each other, a phenomemon known as interference. If the waves are in phase (in step), they reinforce each other. This is called constructive interference. If they are out of phase (out of step), they cancel each other out. This is called destructive interference. Astronomers use the interference between light beams from different parts of stars to image them.

REAL WORLD

Bubble colors

When light is reflected from a soap bubble, some is reflected from the inner surface of the bubble, and some from the outer surface. The light rays from the two surfaces interfere to produce new wavelengths, seen as different colors.



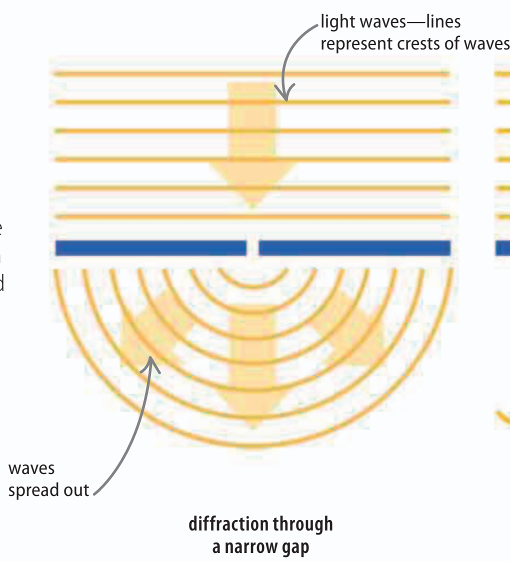


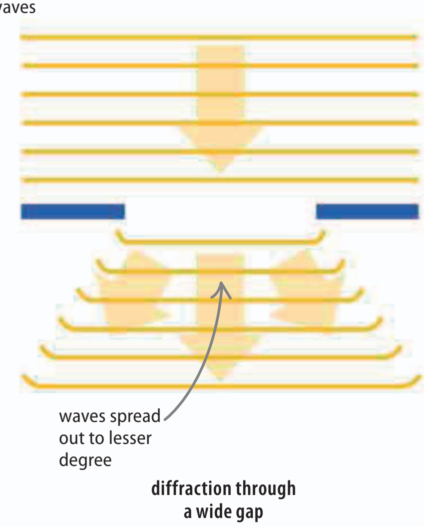
△ **In phase**
When two waves that are in phase meet, their amplitudes add together to make a single wave with double the amplitude. This is called constructive interference.

△ **Out of phase**
If the waves are out of phase, the interference is destructive. As the two waves come together, their amplitudes cancel each other out and the wave is destroyed.

Diffraction

Experiments with light have helped scientists to understand its properties. We know that light can travel as waves, because it behaves like other types of waves, such as sound. For example, both light and sound waves are reflected and refracted (see page 197). Another feature of waves is that, when they pass through a gap, or around an obstacle, they spread out. This effect is called diffraction.





▷ **Spreading out**
Waves spread out like ripples as they pass through a narrow gap. Wider gaps cause less diffraction.